Spectral Transforms

Kauth-Thomas Tasseled Cap Transformation(TCT)

(developed by Kauth and Thomas (1976) Crist and Kauth, 1986)

Kauth-Thomas Transformation

- Sensor systems are often designed to collect data in a larger number of spectral bands than would be needed for any specific application or task, because each possible task is likely to be best served by a different set of bands
- Thus, means are needed by which to determine the best set of bands to use in any given classification circumstance

Kauth-Thomas Transformation

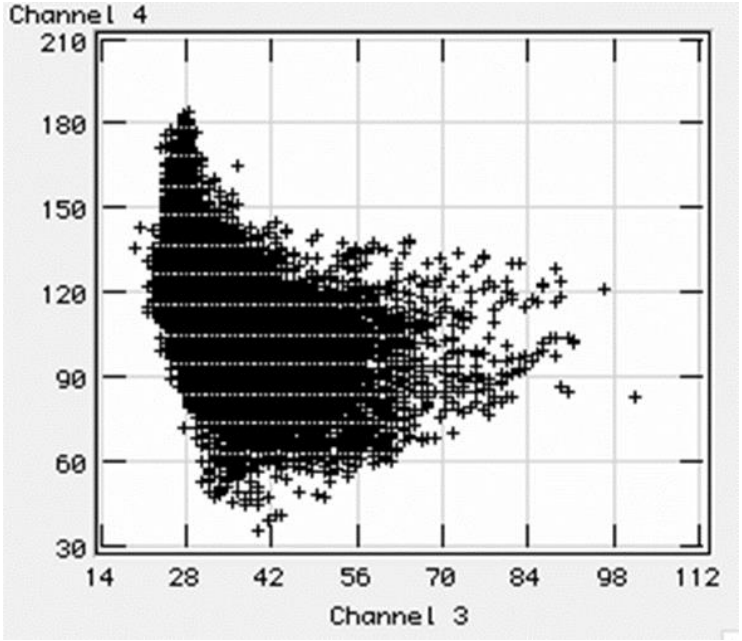
- There can be several different objectives in determining the bands set to use in a specific instance
- Some are useful for image enhancement tasks in order to make certain characteristics of a scene more apparent to a human observer
- Others are intended for use in calculating optimal bands for use with a pattern recognition or classification algorithms

Kauth-Thomas Tasseled Cap Transformation

- Originally developed for Landsat MSS data to represent physical properties of growing crops in Midwest, US
- Provides means to optimize data viewing for vegetation studies
- Sensor dependent
- Can be useful to enhance human perception for the interpretation of a pixel/pixels
- It does not enhance separability

Tasseled Cap Transformation

- The plot of Red and IR reflectance is called a **Tasseled-Cap**, because of its shape

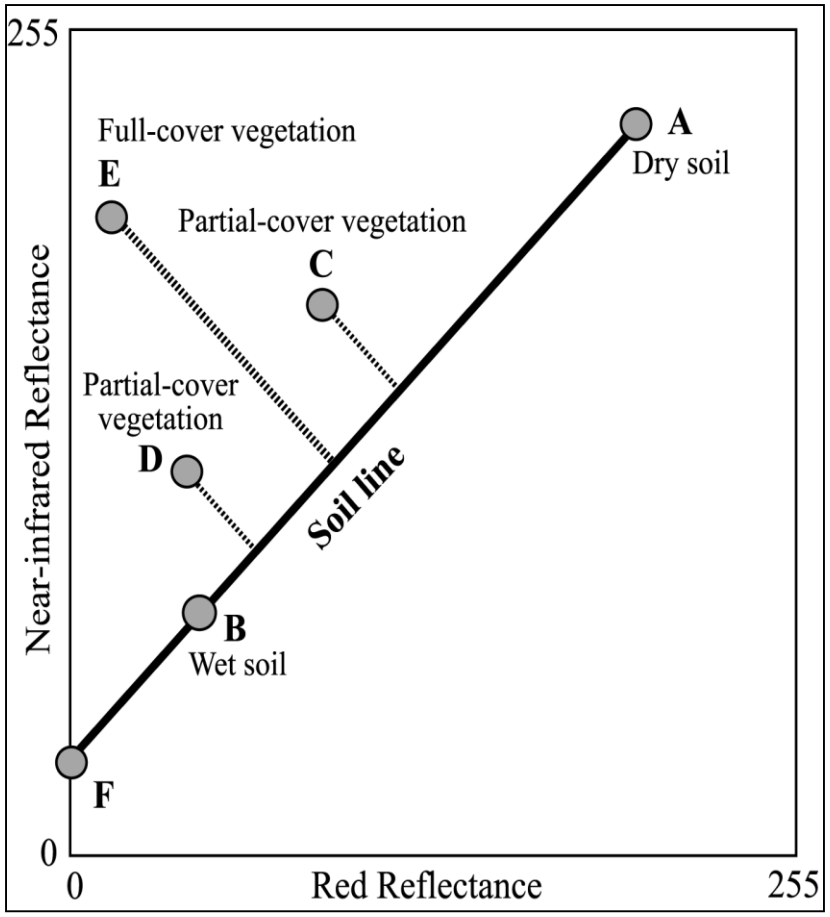




Tasseled Cap Transformation

- In the plot of Red and IR reflectance, perpendicular distance to soil line is an indicator of plant development

- The farther the distance perpendicular to soil line, the greater the amount of vegetation present



Kauth-Thomas Tasseled Cap Transformation

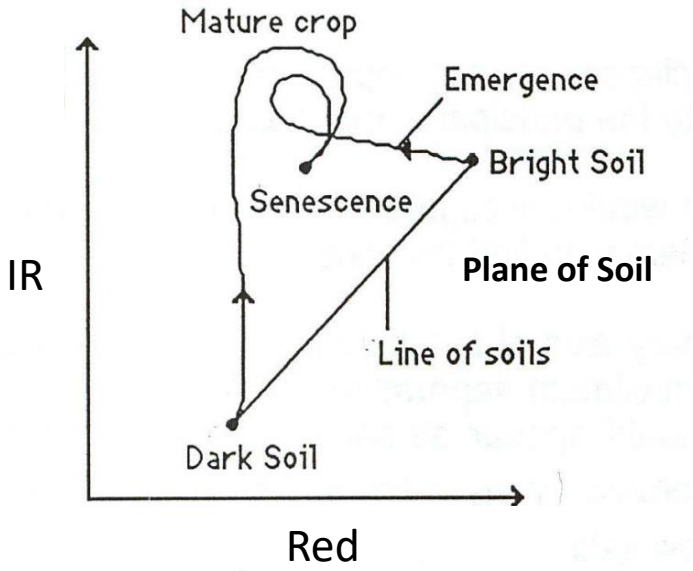
- Axis 1: Soil brightness /reflectance of non- vegetated areas (pre-emergence)
- Axis 2: Greenness: Orthogonal to Axis 1, in direction of vegetation (signature moves up axis as crops grow), vegetation index
- Axis 3: Yellow: Orthogonal to Axes 1 and 2 (as crops senesence, signature increases in yellow direction)
- Axis 4: Wetness (non- such)-related to soil moisture

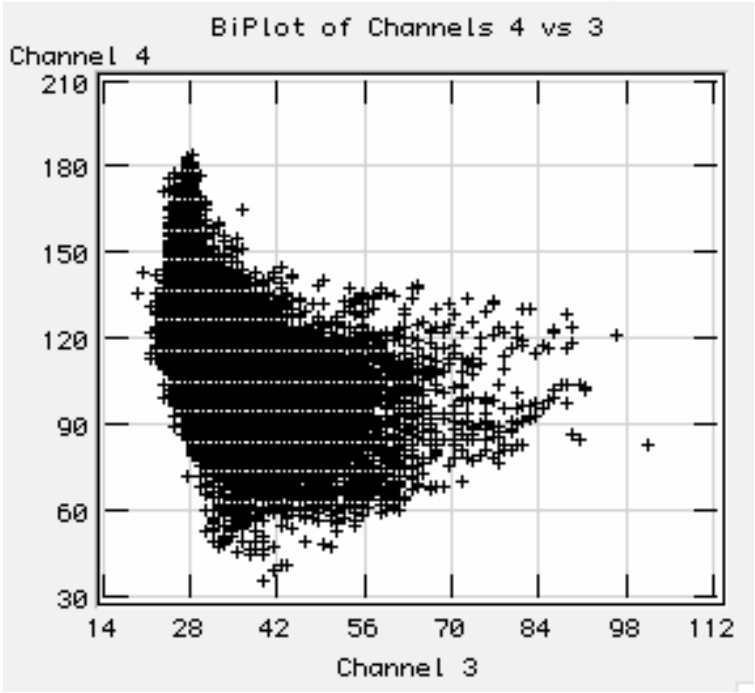
Tasseled Cap Transformation

Transformation identify four new axis:

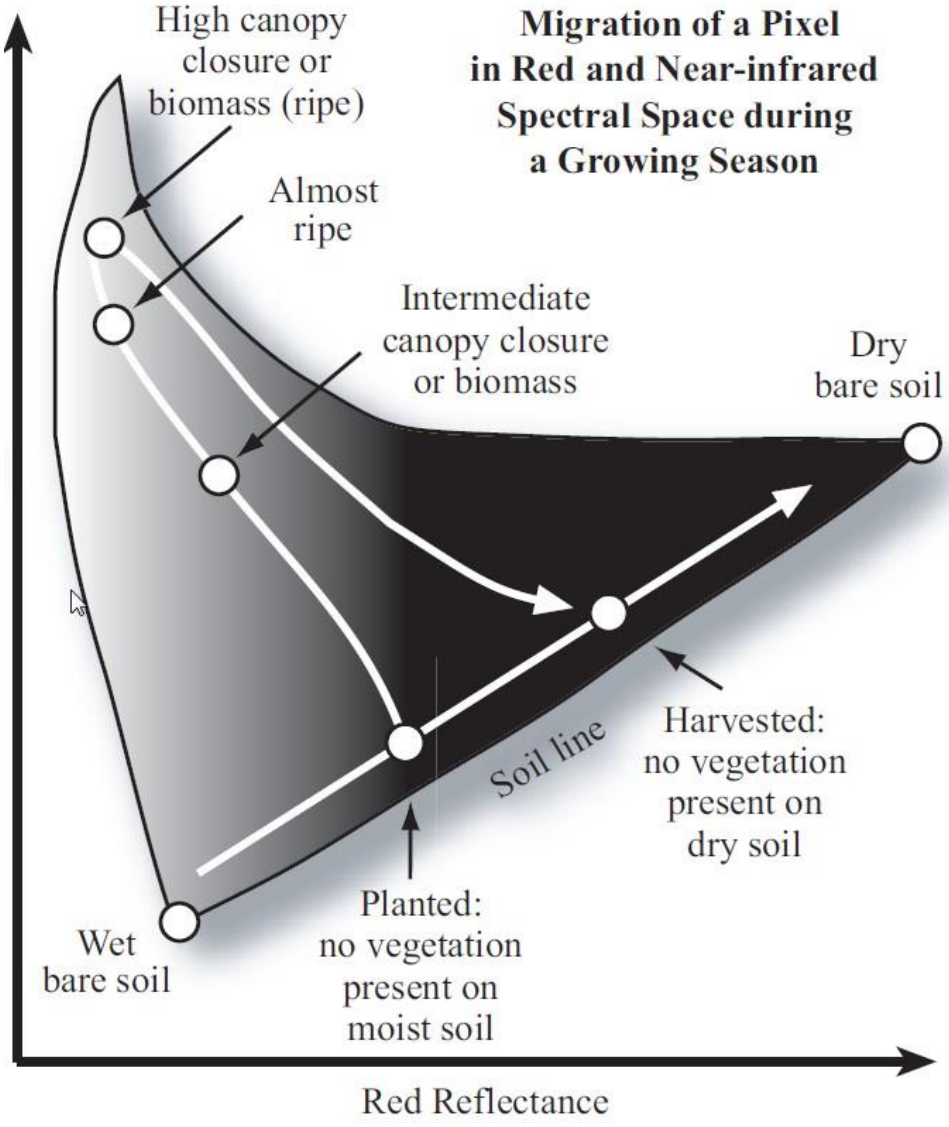
- Brightness -Soil Brightness Index (SBI)
- Greenness – Green Vegetation index (GVI)
- Yellow Stuff Index (YVI)
- Non-Such Index (NSI)







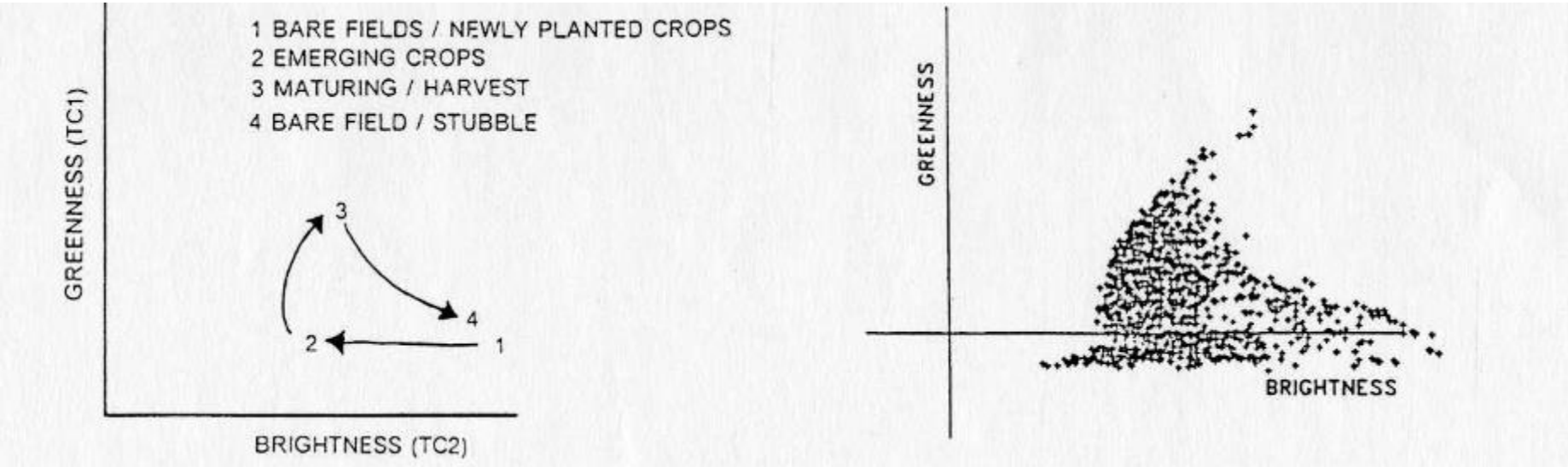
Characteristics of Kauth-Thomas Tasseled Cap Transformation



Tasseled Cap Transformation

The sequence:

1. Bare fields / newly planted crops - high brightness, low greenness (spring)
2. Plant Growth – *increasing greenness and decreasing* brightness (early summer)
3. Maturity: greenness (late summer)
4. Senescence (harvest) - bare field/stubble: *decreasing* greenness, *increasing* brightness (Fall)



Tasseled Cap Transformation

- SBI,GVI and YVI images can be created by multiplying each of the original brightness values of each pixel by corresponding TC Coefficients

$$\mathbf{TC} = \mathbf{W}_{TC} \cdot \mathbf{DN}$$

- TCT has a fixed $\mathbf{W}_{TC}$ and is independent of the scene

Brightness – Greenness - Wetness

- **The transform developed for the MSS consisted of coefficients that extracted brightness and greenness. This was due to the spectral resolution of the MSS that focused primarily in the visible and near infrared**

B = 0.332 MSS1 + 0.603 MSS2 + 0.675 MSS3 + 0.262 MSS4

G = -0.283 MSS1 – 0.660 MSS2 + 0.577 MSS3 + 0.388 MSS4

Brightness – Greenness - Wetness

Brightness -defined in the direction of soil reflectance variations, urbanized and bare soil areas are evident in this image

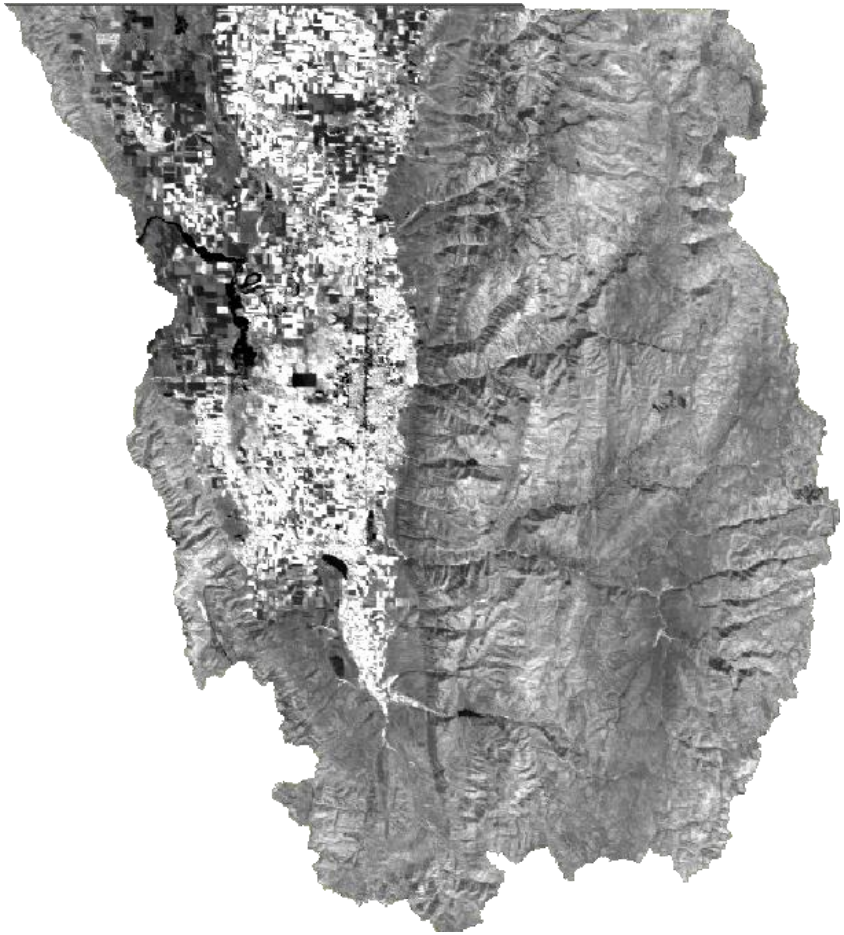




Brightness – Greenness - Wetness

Greenness -defined in the direction of vegetation reflectance variations, the greater the biomass, the brighter the pixel value in this image

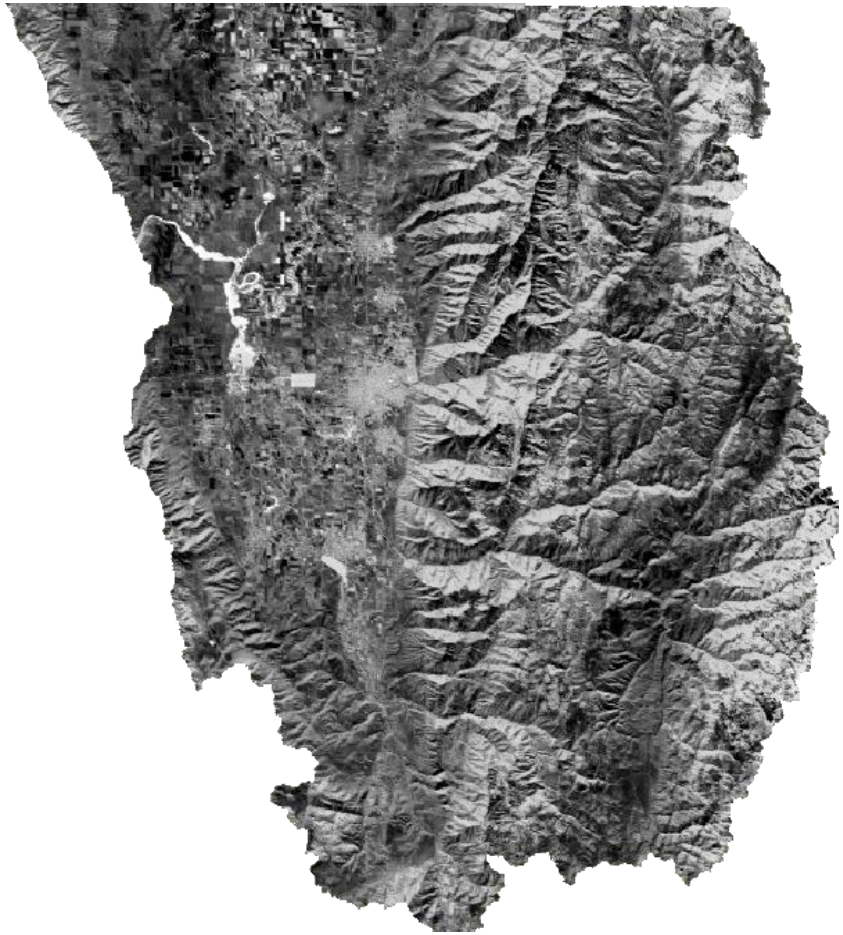




Brightness – Greenness - Wetness

Wetness - information concerning soil & plant moisture
water bodies are very bright greater the moisture content, brighter response





Brightness – Greenness - Wetness

- **The coefficients were recalculated to take advantage of the increased spectral resolution of the TM**
- **This allowed for the extraction of an additional component called wetness due to the inclusion of the MIR channels that are sensitive to moisture**

B = 0.2909 TM1 + 0.2493 TM2 + 0.4806 TM3 + 0.5568 TM4 + 0.4438 TM5 +0.1706 TM7

G = -0.2728 TM1 – 0.2174 TM2 - 0.55508 TM3 + 0.7221 TM4 + 0.0733 TM5 – 0.1648 TM7

W = 0.1446 TM1 + 0.1761 TM2 + 0.3322 TM3 + 0.3396 TM4 -0.6210 TM5 – 0.4186 TM7

Tasseled Cap Coefficients

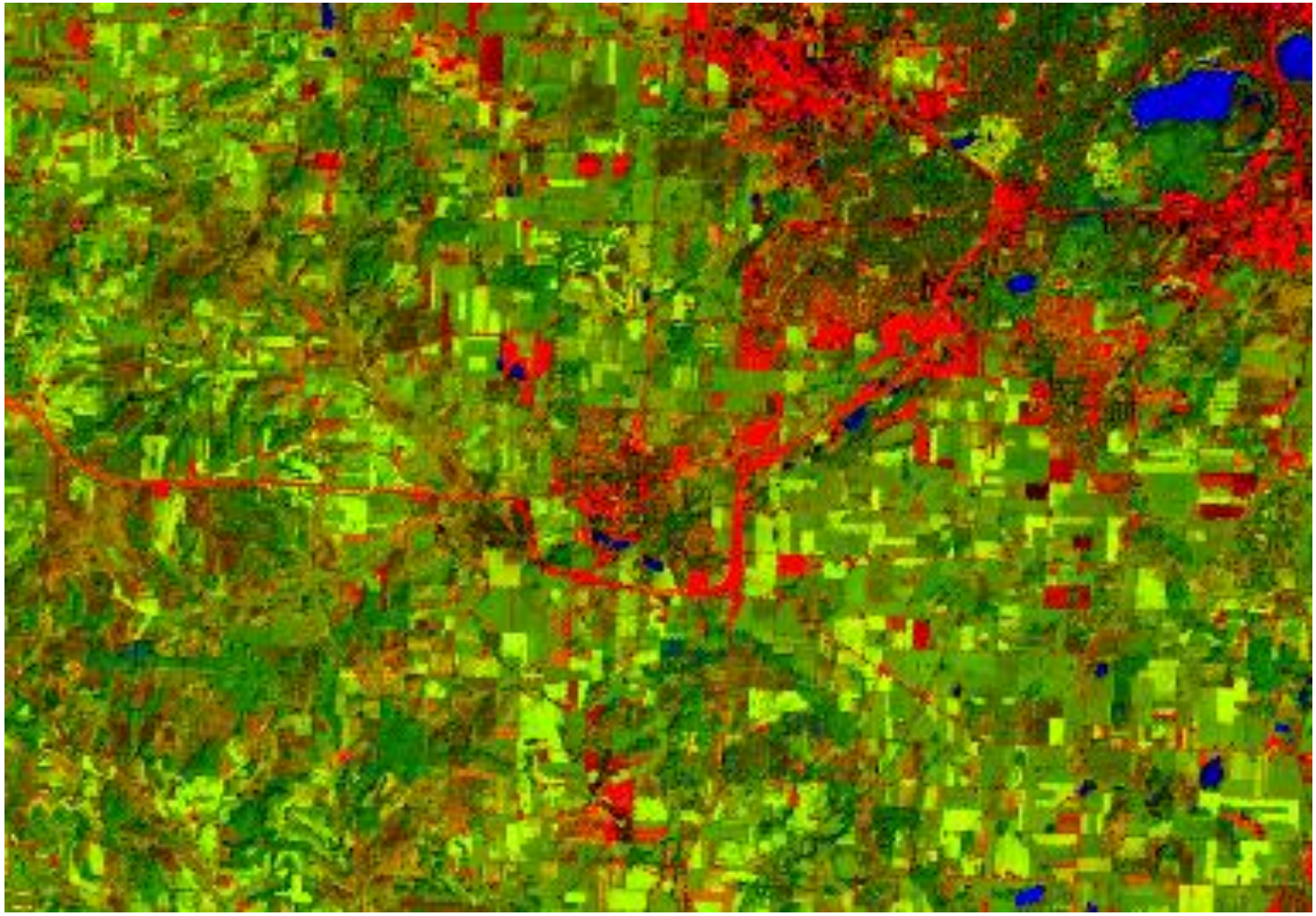
sensor	axis name	W_{TC}
L-7	soil brightness	$\begin{bmatrix} +0.3561 & +0.3972 & +0.3904 & +0.6966 & +0.2286 & +0.1596 \\ -0.3344 & -0.3544 & -0.4556 & +0.6966 & -0.0242 & -0.2630 \\ +0.2626 & +0.2141 & +0.0926 & +0.0656 & -0.7629 & -0.5388 \end{bmatrix}$
	greeness	
	wetness	

Landsat 8 / TCT	(Blue) Band 2	(Green) Band 3	(Red) Band 4	(NIR) Band 5	(SWIR1) Band 6	(SWIR2) Band 7
Brightness	0.3029	0.2786	0.4733	0.5599	0.508	0.1872
Greenness	−0.2941	−0.243	−0.5424	0.7276	0.0713	−0.1608
Wetness	0.1511	0.1973	0.3283	0.3407	−0.7117	−0.4559

Muhammad Hasan Ali Baig, Lifu Zhang, Tong Shuai & Qingxi Tong (2014), Derivation of a tasselled cap transformation based on Landsat 8 at-satellite reflectance, Remote, Sensing Letters, 5:5, 423-431, DOI: 10.1080/2150704X.2014.915434

http://dx.doi.org/10.1080/2150704X.2014.915434

Tasseled Cap Transformation



Red- Brightness Green-Greenness Blue- Wetness

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